DISPROOF OF TERMINATION IN ERDŐS PROBLEM 282 (GRAHAM'S VARIANT)

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For $A \subseteq \mathbb{N}$ and $x \in \mathbb{Q}$, define the greedy unit-fraction remainder sequence $\mathrm{rem}(A, x, t) \in \mathbb{Q}$ ($t \in \mathbb{N}$) recursively by $\mathrm{rem}(A, x, t) := 0$ when $x \le 0$, and for $x > 0$ by letting $n := \inf\{k \in \mathbb{N} : k \in A \text{ and } 1/x \le k\}$, setting $r := x - 1/n$, and putting

$$\mathrm{rem}(A, x, 0) := r, \qquad \mathrm{rem}(A, x, t+1) := \mathrm{rem}(A \setminus \{n\}, r, t).$$

Graham [1] proved that a positive rational m/n is a finite sum of distinct unit fractions with denominators $\equiv a \pmod d$ ($d > 1$) if and only if $\gcd\big(n/\gcd(n, a, d),\, d/\gcd(a, d)\big) = 1$. Erdős and Graham [2, p. 31] (see [3, 4]) asked whether the greedy algorithm restricted to $\{n \in \mathbb{N} : n \equiv a \pmod d\}$ always terminates whenever this condition holds.

Theorem. *For $x = 1/119 \in (0, 1)$, $a = 1$, and $d = 3$, Graham's coprimality condition holds ($\gcd(119, 3) = 1$), yet $\mathrm{rem}(\{n \in \mathbb{N} : n \equiv 1 \pmod 3\}, 1/119, \cdot)$ is not eventually zero.*

Correspondence with the Lean formalization. The proof below is the one formalized in `FormalConjectures/ErdosProblems/282.lean`, step by step, with the same names (m, m', q, q', c, K, u_2, q_3, v_3, u_4, k_5, t, c', T) and with every identity and inequality in the form used in Lean. All variables are natural numbers unless noted otherwise. The correspondences are:

This note	Lean
$\mathrm{rem}(A, x, t)$	`greedyUnitFractionRem A x t`
$\mathrm{ValidSet}(A, q)$	`ValidSet A q`
$\mathrm{sFrac}(m, q) := m/(3mq-1) \in \mathbb{Q}$	`sFrac m q`
Lemma 1	`greedy_step`
Lemma 2	`pos_at_six_mul`
$u_2, q_3, v_3, u_4, k_5, t, c'$	`u2`, `q3`, `v3`, `u4`, `k5`, `t`, `c'`
Theorem	`erdos_282.variants.graham`

For $A \subseteq \mathbb{N}$ and $q \in \mathbb{N}$, define the predicate $\mathrm{ValidSet}(A, q)$ to mean that $1 \le q$ and for every $k \in \mathbb{N}$:

$$(k \in A \Rightarrow k \bmod 3 = 1) \qquad \text{and} \qquad (3q < k \text{ and } k \bmod 3 = 1 \Rightarrow k \in A).$$

For $m, q \in \mathbb{N}$, put $\mathrm{sFrac}(m, q) := m/(3mq-1) \in \mathbb{Q}$.

Lemma 1. *Let $m, m', q, q' \in \mathbb{N}$ and $A \subseteq \mathbb{N}$ satisfy $1 \le m$, $1 \le m'$, $\mathrm{ValidSet}(A, q)$, $q < q'$, and*

$$(m+1)(3m'q' - 1) = m'(3mq-1)(3q+1) \qquad \text{in } \mathbb{Q}. \tag{1}$$

Then:

(i) $\mathrm{ValidSet}(A \setminus \{3q+1\}, q')$;
(ii) $\mathrm{rem}(A, \mathrm{sFrac}(m, q), 0) = \mathrm{sFrac}(m', q')$;
(iii) $\mathrm{rem}(A, \mathrm{sFrac}(m, q), t+1) = \mathrm{rem}(A \setminus \{3q+1\}, \mathrm{sFrac}(m', q'), t)$ *for all $t \in \mathbb{N}$.*

Proof. Since $1 \le m$ and $1 \le q$, we have $3mq - 1 > 0$, so $\mathrm{sFrac}(m, q) > 0$. Set $S := \{k \in \mathbb{N} : k \in A \text{ and } 1/\mathrm{sFrac}(m, q) \le k\}$. Because $3q < 3q+1$ and $(3q+1) \bmod 3 = 1$, $\mathrm{ValidSet}(A, q)$ gives $3q+1 \in A$, and $3mq - 1 \le (3q+1)m$ gives $1/\mathrm{sFrac}(m, q) \le 3q+1$, so $3q+1 \in S$. Conversely, let $k \in S$. Then $3mq \le km + 1$. If $k < 3q+1$, then $k \bmod 3 = 1$ (from $k \in A$) forces $k+2 \le 3q$, so $(k+2)m \le 3mq \le km + 1$, contradicting $1 \le m$. Hence $3q + 1 \le k$ for all $k \in S$, which proves $\inf S = 3q + 1$.

Next, clearing denominators against $(3mq-1)(3q+1) \ne 0$ and $3m'q' - 1 \ne 0$, the identity $\mathrm{sFrac}(m, q) - 1/(3q+1) = \mathrm{sFrac}(m', q')$ reduces to

$$\big(m(3q+1) - (3mq-1)\big)(3m'q' - 1) = m'(3mq-1)(3q+1),$$

which holds by (1) since $m(3q+1)-(3mq-1)=m+1$. Finally, $\mathrm{ValidSet}(A\setminus\{3q+1\},q')$ holds because $1\le q<q'$ implies $1\le q'$, membership in $A\setminus\{3q+1\}$ implies membership in A (hence $k \bmod 3=1$), and any $k>3q'$ with $k \bmod 3=1$ satisfies $k>3q+1$ (so $k\in A$ and $k\neq 3q+1$). Unfolding rem at $\mathrm{sFrac}(m,q)>0$ now yields (ii) and (iii). □

Lemma 2. *For every $K\in\mathbb{N}$, $A\subseteq\mathbb{N}$, and $c\in\mathbb{N}$ satisfying* $\mathrm{ValidSet}(A,2c)$, *$c \bmod 7=6$, and ($c \bmod 5=0$ or $c \bmod 25=18$), we have*

$$0<\mathrm{rem}(A,\mathrm{sFrac}(1,2c),6K).$$

Proof. We induct on K, quantifying over A and c. Note first that $1\le 2c$ implies $1\le c$.

Base case ($K=0$). Applying Lemma 1 (ii) with $(m,m',q,q')=(1,2,2c,6c^2)$ (since $2c<6c^2$ for $1\le c$ and $2(36c^2-1)=2(6c-1)(6c+1)$) gives $\mathrm{rem}(A,\mathrm{sFrac}(1,2c),0)=2/(36c^2-1)>0$.

Inductive step ($K\to K+1$). Set $u_2:=c^2(36c^2+1)$ and $q_3:=u_2(9u_2+1)$. Regardless of the parity of u_2, $u_2(9u_2+1)$ is even, so $q_3 \bmod 2=0$. Modulo 25:

- if $c \bmod 5=0$, writing $c=5r$ gives $25\mid c^2\mid u_2\mid q_3$, so $q_3 \bmod 25=0$;
- if $c \bmod 25=18$, evaluating u_2 and q_3 modulo 25 gives $q_3 \bmod 25=10$.

In either case $10\mid q_3$, so write $q_3=10v_3$. Set $u_4:=v_3(40v_3+1)$ and $k_5:=u_4(45u_4+2)$. From $c \bmod 7=6$ we compute $q_3 \bmod 7=3$, hence $v_3 \bmod 7=1$ (since $10v_3=q_3$), hence $u_4 \bmod 7=6$ and $k_5 \bmod 7=1$. Writing $t:=\lfloor k_5/7\rfloor$ (so $k_5=7t+1$) and setting

$$c':=7(36t^2+11t)+6,$$

ring expansion gives the exact identity

$$7c'=36k_5^2+5k_5+1, \tag{2}$$

and visibly $c' \bmod 7=6$. For the modulo-5/25 disjunction on c', note from $10v_3=q_3$ and $q_3 \bmod 25\in\{0,10\}$ that $v_3 \bmod 5\in\{0,1\}$:

- if $v_3 \bmod 5=0$, then $5\mid v_3\mid u_4\mid k_5$; writing $k_5=5r$ in (2) gives $7c'=25(36r^2+r)+1$, whence $c' \bmod 25=18$;
- if $v_3 \bmod 5=1$, then $u_4 \bmod 5=1$ and $k_5 \bmod 5=2$, so (2) gives $(7c') \bmod 5=(36\cdot 4+10+1) \bmod 5=0$, whence $c' \bmod 5=0$.

Now apply Lemma 1 six times starting from $A_0:=A$:

(1) $(m,m',q,q')=(1,2,2c,6c^2)$: $2c<6c^2$ since $1\le c$, and $2(36c^2-1)=2(6c-1)(6c+1)$;
(2) $(m,m',q,q')=(2,3,6c^2,2u_2)$: $6c^2<2c^2(36c^2+1)$ since $1\le c^2$, and $3(18u_2-1)=3(36c^2-1)(18c^2+1)$;
(3) $(m,m',q,q')=(3,4,2u_2,q_3)$: $2u_2<u_2(9u_2+1)$ since $1\le 2u_2$, and $4(12q_3-1)=4(18u_2-1)(6u_2+1)$;
(4) $(m,m',q,q')=(4,5,q_3,6u_4)$: using $q_3=10v_3$, $10v_3<6v_3(40v_3+1)$ since $1\le q_3$, and $5(90u_4-1)=5(120v_3-1)(30v_3+1)$;
(5) $(m,m',q,q')=(5,6,6u_4,2k_5)$: $6u_4<2u_4(45u_4+2)$ since $1\le 6u_4$, and $6(36k_5-1)=6(90u_4-1)(18u_4+1)$;
(6) $(m,m',q,q')=(6,1,2k_5,2c')$: using (2), $2k_5<2c'$ since $1\le 2k_5$, and $7(6c'-1)=(36k_5-1)(6k_5+1)$ since $7(6c'-1)=6(7c')-7=6(36k_5^2+5k_5+1)-7=216k_5^2+30k_5-1=(36k_5-1)(6k_5+1)$.

Writing $6(K+1)=6K+1+1+1+1+1+1$, the six step equations (iii) rewrite the remainder $\mathrm{rem}(A,\mathrm{sFrac}(1,2c),6(K+1))$ to $\mathrm{rem}(A_6,\mathrm{sFrac}(1,2c'),6K)$ with $\mathrm{ValidSet}(A_6,2c')$, which is positive by the induction hypothesis applied to A_6 and c'. □

Proof of the Theorem. Take $x=1/119\in(0,1)$, $a=1$, and $d=3$; then $\gcd(1,3)=1$ and $\gcd(119,3)=1$. Suppose for contradiction that $\mathrm{rem}(A,1/119,t)=0$ for all $t\ge T$, where $A:=\{n\in\mathbb{N}:n\equiv 1 \pmod 3\}$. For $c=20$, we have $\mathrm{ValidSet}(A,40)$, $20 \bmod 7=6$, $20 \bmod 5=0$, and $\mathrm{sFrac}(1,40)=1/119$. Lemma 2 with $K=T$ gives $0<\mathrm{rem}(A,1/119,6T)$, contradicting $\mathrm{rem}(A,1/119,6T)=0$ since $T\le 6T$. □

References

[1] R. L. Graham, *On finite sums of unit fractions*, Proc. London Math. Soc. (3) **14** (1964), 193–207.
[2] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monographies de L'Enseignement Mathématique, vol. 28, Université de Genève, 1980.
[3] *Erdős Problem #282*, `https://www.erdosproblems.com/282`.
[4] The Formal Conjectures Authors, *FormalConjectures/ErdosProblems/282.lean*, `https://github.com/google-deepmind/formal-conjectures`.